

Linear Regression Analysis - Predicting Body Mass Index

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1 Introduction to Linear Modeling

The goal is to explain which factors are associated with BMI in US adults (NHANES dataset), controlling for demographics (age, gender, race, education), socio-economic indicators (education, income) and lifestyle (sleep, physical activity, alcohol, smoking). We will start simple and incrementally extend to a multiple linear regression, also adding multiple regression effects that are conditional on the other covariates in the model.

1.1 Linear Models for BMI

Body Mass Index (BMI) is a continuous variable calculated as weight (kg) divided by height squared (m^2). BMI is a proxy for body fat and is strongly related to chronic diseases such as diabetes, cardiovascular disease and hypertension.

In this analysis, we use NHANES adult participants (Age ≥ 18) to examine how demographics, socioeconomic status and lifestyle behaviors are associated with BMI

1.2 Model Specifications

We model BMI as a linear function of selected predictors:

$$\text{BMI} = \beta_0 + \beta_1 * \text{Age} + \beta_2 * \text{Gender} + \beta_3 * \text{Race} + \beta_4 * \text{Education} + \beta_5 * \log(\text{Income}) + \beta_6 * \text{PhysActive} + \beta_7 * \text{SleepHrs} + \beta_8 * \text{SmokeNow} + \beta_9 * \text{AlcoholDay} + \epsilon$$

Where:

- β_0 is the intercept (BMI)
- $\beta_1 \dots \beta_8$ are the regression coefficients representing the effect on each predictor.
- ϵ represents the random error term (assumed that it is normally distributed)

1.3 Research Questions

- After adjusting for covariates, how does BMI vary with Age?
- Do demographic factors (Gender, Race, Education) show overall association with BMI?
- Are lifestyle factors (PhysActive, AlcoholDay, SleepNight, SmokeNow) associated with BMI, and how much?
- How much variance is explained by the model?

2 Data processing

- Population: NHANES adults (Age ≥ 18).
- Variables: Age; Gender; Race1; Education; HHIncomeMid (we used log transform); PhysActive; SleepHrsNight; AlcoholDay; SmokeNow; BPSysAve. Factor coding: treatment contrasts (reference vs others).
- Missing data strategy (baseline): Complete-case analysis on these variables to keep the workflow transparent.

i Handling Missing Values

We included only adults (≥ 18 years). BMI in children is interpreted with age- and sex-specific percentiles, so combining adults and minors would yield non-comparable BMI categories and biased estimates. We created $\log_income = \log(\text{HHIncomeMid})$. We then used a complete-case dataset for baseline modeling (all variables observed), retaining 27.5% of the adult sample.

AlcoholDay/SmokeNow are driving most of the loss.

Table 2.1: Percentage of Missing Values

Variable	Percent_Missing
SmokeNow	57.1
AlcoholDay	34.3
HHIncomeMid	8.6
$\log_income$	8.6
BPSysAve	3.7
Education	3.5
BMI	0.9
SleepHrsNight	0.2
Age	0.0
Race1	0.0
Gender	0.0
PhysActive	0.0

3 Exploratory Analysis (EDA)

3.1 Outcome distribution (BMI)

Summary statistics: mean, median, SD, quantiles:

The response variable, Body Mass Index (BMI), ranged from 15.0 to 81.2 kg/m² with a mean of 28.8 kg/m² (SD = 6.65, median = 27.8).

Approximately 33% of participants were classified as overweight (25 ≤ BMI < 30) and 33% as obese (BMI ≥ 30), reflecting the high prevalence of excess weight in the U.S. adult population.

The distribution exhibited moderate right skewness (skewness = 1.2), indicating a longer tail with high BMI values

Table 3.1: Descriptive Statistics for BMI

Mean	Median	SD	Min	Max	Q1	Q3	IQR	Skewness
28.3	27.3	6.2	15	67.8	24	31.6	7.6	1.2

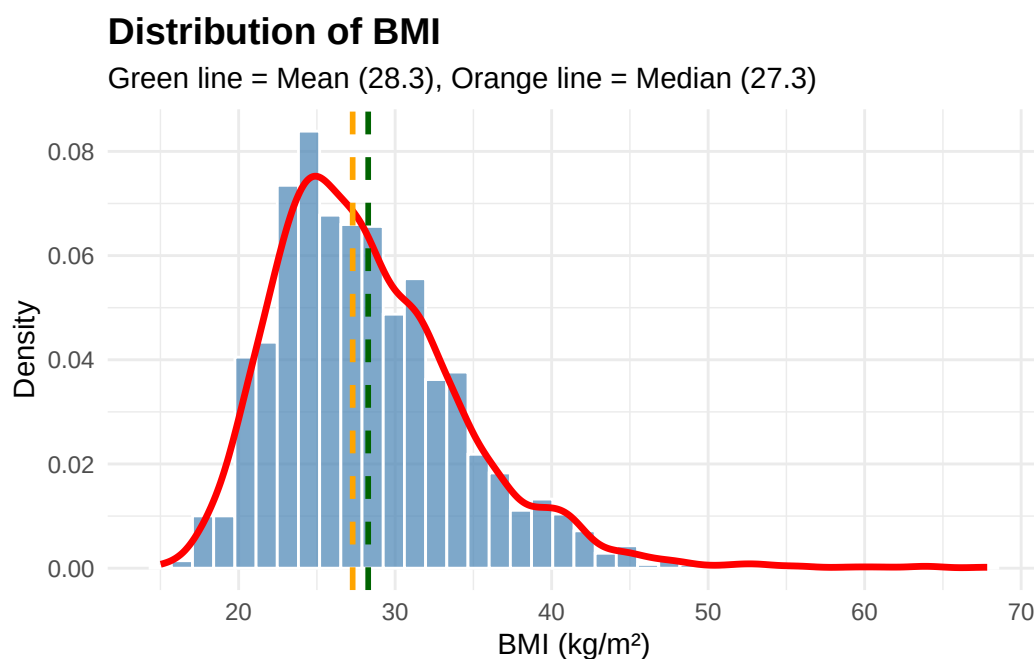
Table 3.2: Distribution of BMI Categories

BMI Category	Patient Count	Percentage (%)
Underweight	35	1.7
Normal	652	31.7
Overweight	687	33.3
Obese	686	33.3

i BMI Distribution plot and outlier detection

Distribution plot of BMI (histogram + density curve)

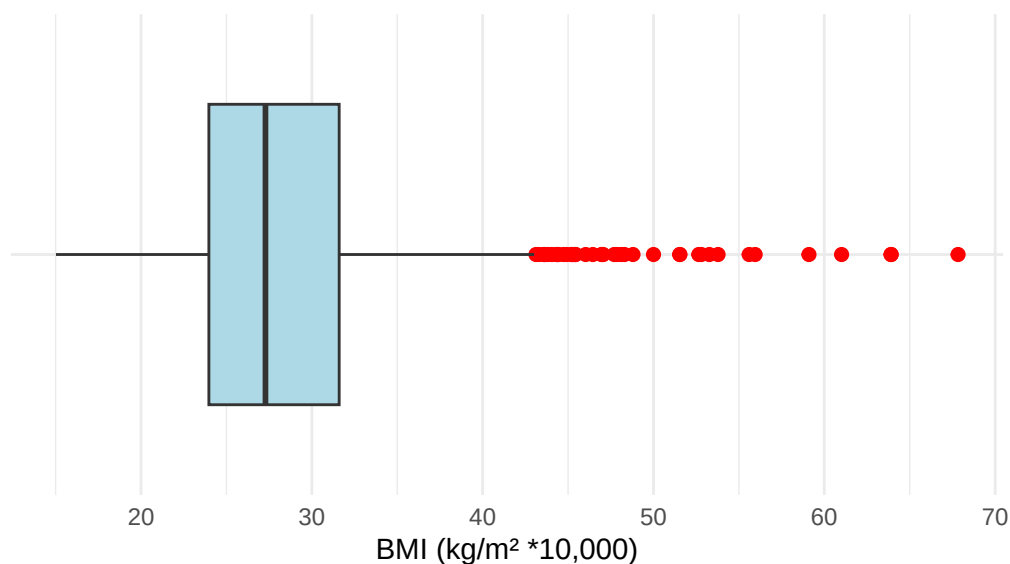
The distribution of the BMI variable is right-skewed as shown in the overview. This shows that most people are around the average BMI in the data; however, some have very high BMI



The box plot indicates that there are no outliers in the lower part of the distribution. The lower threshold is at approximately 13, while the upper threshold is approximately at 43. In contrast the upper tail displays a substantial number of extreme values, with 43 observations identified as outliers.

Boxplot of BMI (Outlier Detection)

Red points are statistical outliers ($> 1.5 \times \text{IQR}$ from quartiles)



Although the BMI distribution shows high-value observations, these values fall within plausible physiological ranges for the NHANES population. Therefore, no outliers were removed.

3.2 Summary of Exploratory Data Analysis BMI vs predictors

Among all variables examined, physical activity, race/ethnicity, and education level showed the strongest associations with BMI. Physically active individuals had noticeably lower BMI

on average, and several race and education groups displayed meaningful mean differences. In contrast, most continuous predictors such as age, income, sleep hours, blood pressure, and alcohol use—showed very weak correlations and offered limited linear explanatory power. Please see below in the collapsed section for the full EDA.

i Pairwise relationships with BMI

3.3 Pairwise relationships with BMI (continuous predictors)

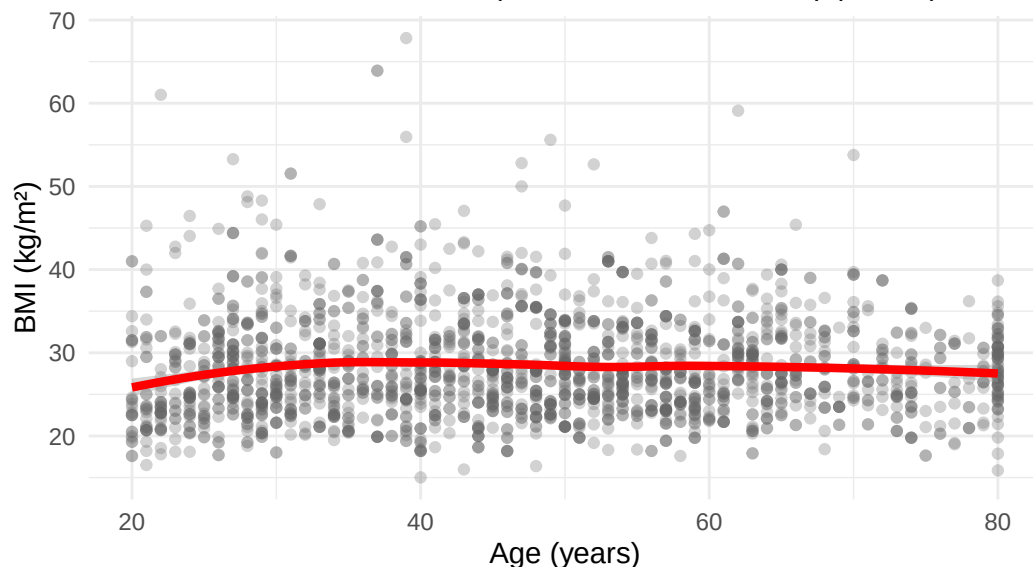
BMI vs Age

The scatterplot with a LOESS smoother shows that BMI remains largely consistent across age groups. The Pearson correlation coefficient ($r = 0.0144721$) indicates virtually no linear association between age and BMI.

The corresponding coefficient of determination ($R^2 = 0.0002094$) confirms that age explains less than 0.02% of the variance in BMI. This suggests that BMI is not influenced by age in this sample, and other demographic or lifestyle variables likely play a more substantial role in determining BMI.

BMI vs Age

Red curve = LOESS smoother (shows actual relationship pattern)



BMI vs log(income)

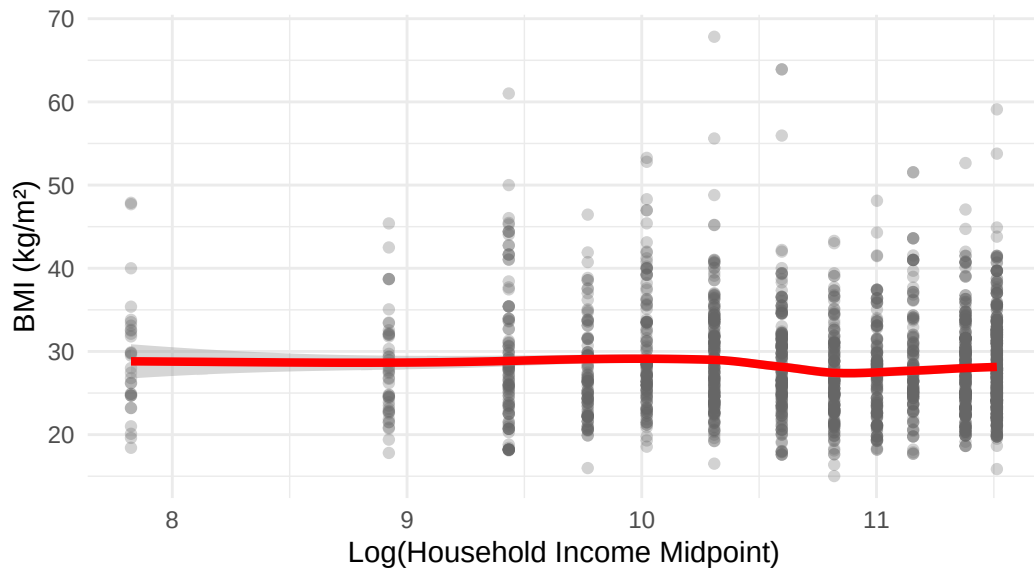
The scatterplot with a LOESS smoother shows a weak negative association between BMI and the logarithm of household income.

The Pearson correlation coefficient ($r = -0.0504988$) confirms that higher income is associated with slightly lower BMI values. However, this relationship is very weak ($R^2 = 0.00255$), indicating that household income explains less than 1% of the variance in BMI.

Although the direction aligns with the expected negative relationship for higher income, the effect size suggests that income has minimal influence on BMI in this sample.

BMI vs Log(Household Income)

Expectation: Negative relationship (higher income → lower BMI)

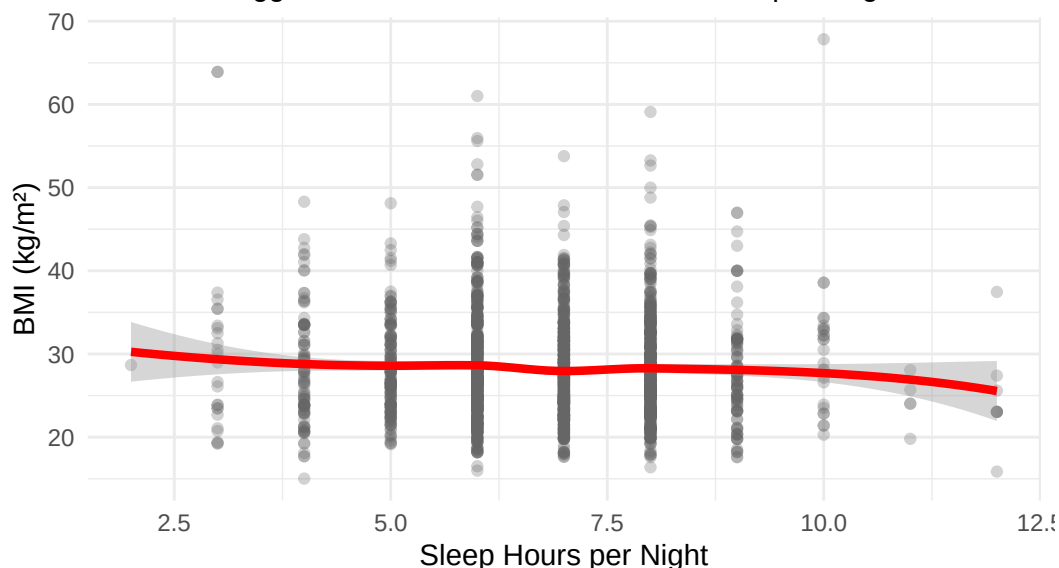


BMI vs Sleep

BMI shows a negligible linear association with sleep duration ($r = -0.0321021$; $R^2 = 0.001031$). The LOESS smoother suggests a shallow U-shape, with the lowest BMI at approximately 7.5 hours of sleep and slightly higher BMI at both shorter and longer durations.

BMI vs Sleep Hours per Night

Research suggests: Both too little and too much sleep → higher BMI



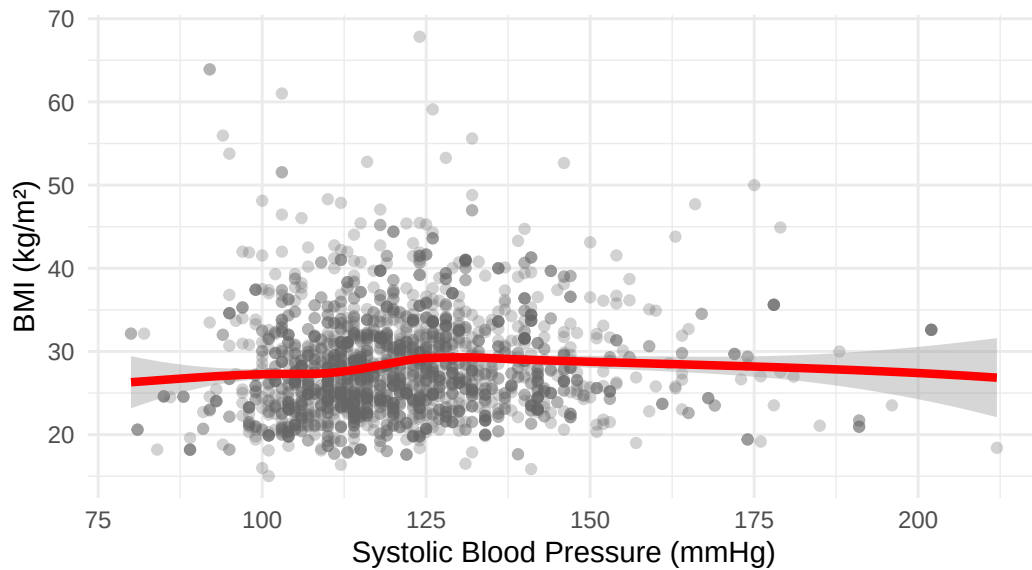
BMI vs Systolic Blood Pressure

Although we expected a strong positive relationship between BMI and systolic blood pressure, the data shows only a very weak positive trend. The LOESS curve suggests a small increase in BP with BMI initially, but then the relationship plateaus and even goes down slightly. This indicates that systolic blood pressure alone is not a predictor in this sample.

BMI is expected to predict high blood pressure, but the data may not show this since many patients manage their blood pressure with medication.

BMI vs Systolic Blood Pressure

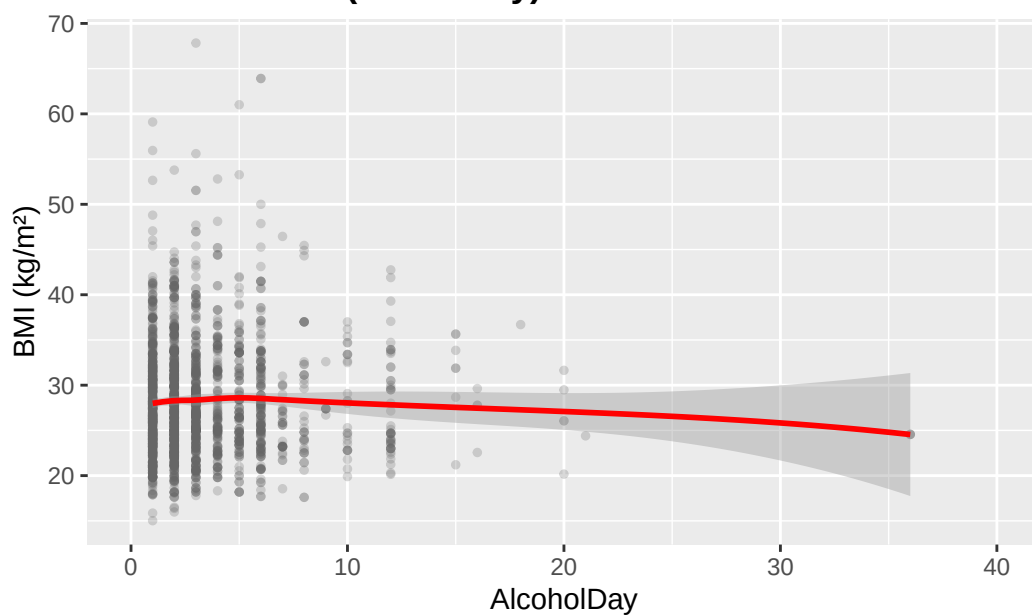
Expectation: higher BMI – higher BP



BMI vs AlcoholDay

The LOESS curve is nearly flat with a slight downward tilt. The confidence interval widens as AlcoholDay increases (due to few observations with higher values). The unadjusted linear correlation is approx 0.03, meaning very little to no association with BMI.

BMI vs Alcohol (drinks/day)



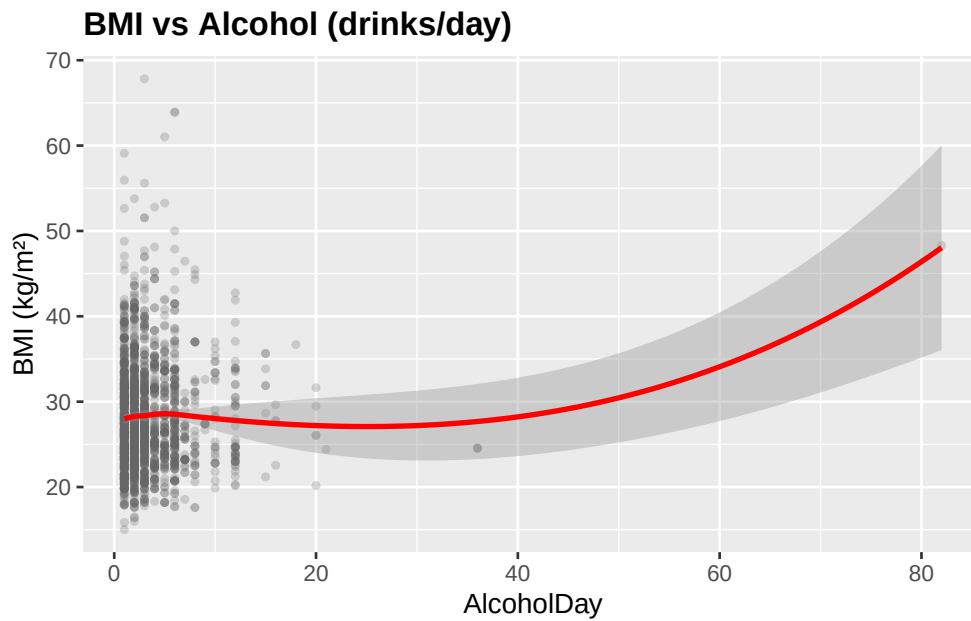
Appendinx: BMI vs Alcohol Day

Below are the BMI–AlcoholDay correlations and visualizations: (i) the plot over the full 0–80 range, and (ii) the $\log(1 + \text{AlcoholDay})$

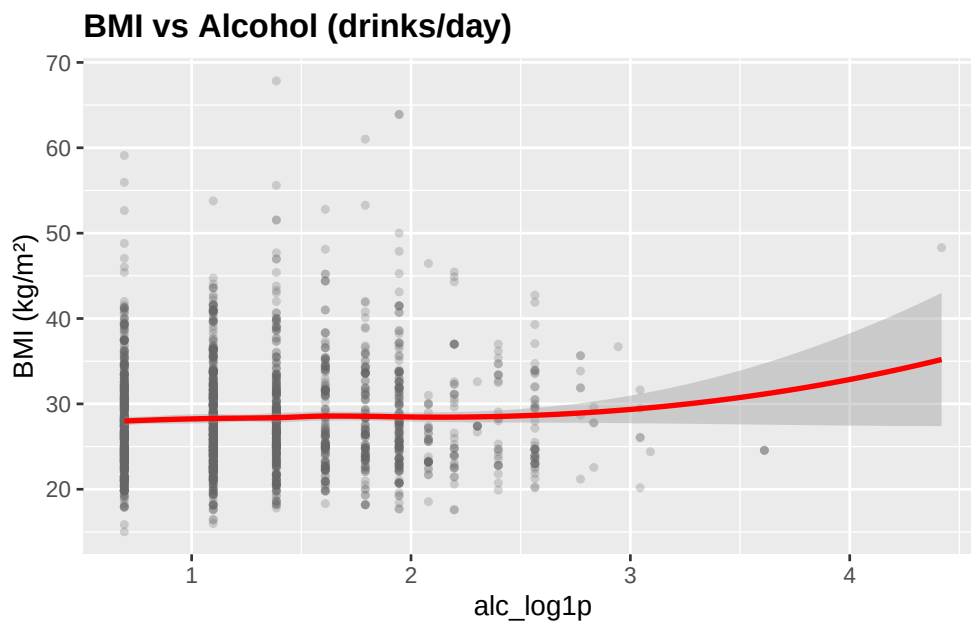
```
cor_alc <- cor(nhanes_lm$AlcoholDay, nhanes_lm$BMI)
cor_alc
```

```
[1] 0.03483019
```

(i) the plot over the full 0–80 range



(ii) the $\log(1 + \text{AlcoholDay})$



[1] 0.0240489

The correlation between alcohol and BMI is even smaller when AlcoholDay is logarithmized - 0.0240489.

3.4 BMI vs Categorical predictors

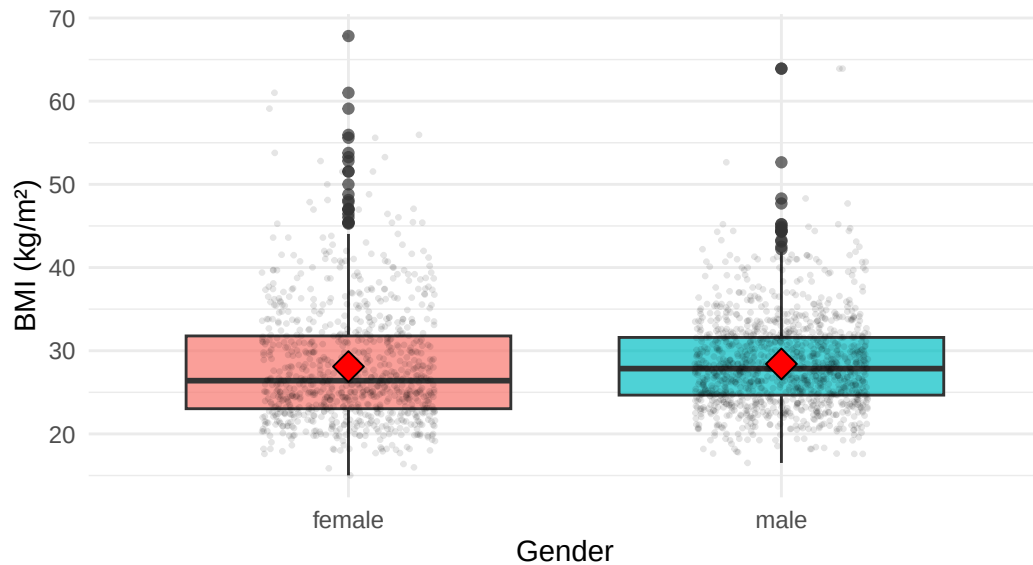
BMI by Gender

In this sample the average for female and male is almost the same; however, the BMI distribution is also more variable among females, as indicated by a higher standard deviation

(sd = 7.04 vs. 5.43). There is *no evidence* that gender plays a strong role in explaining BMI differences in this sample.

BMI by Gender

Red diamond = mean, Box shows median and quartiles

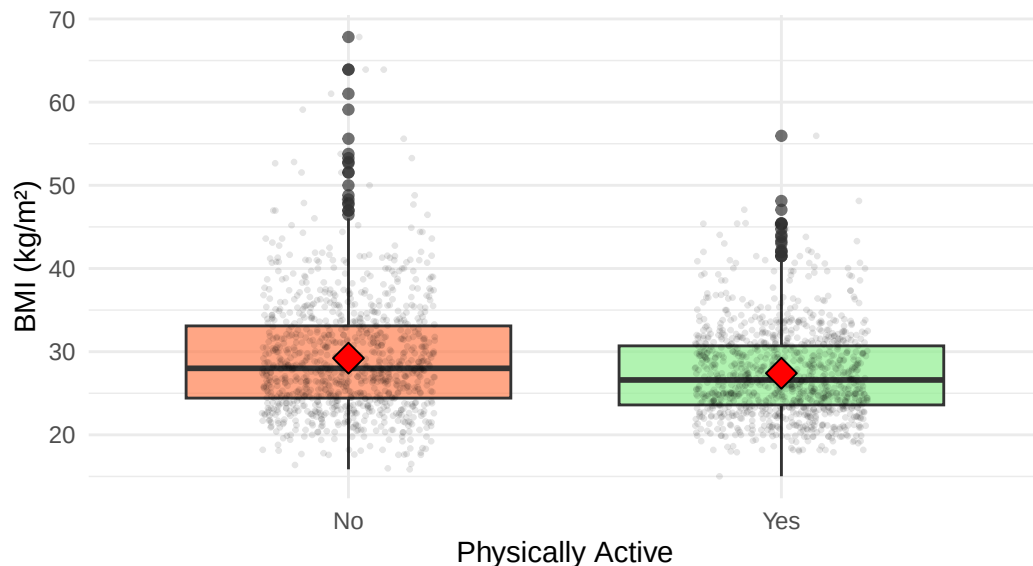


BMI by Physical Activity

On average, individuals that reported being physically active have a lower BMI (mean 27.9 kg/m²) than those who are not (mean 29.9 kg/m²). There is very strong evidence for a difference in BMI between physically active and inactive individuals ($p < 2.2 \times 10^{-16}$), with an estimated mean difference of approximately 2.0 units (95% CI: [1.68, 2.36] kg/m²). BMI is also more variable among inactive individuals (SD = 6.9 vs. 5.3), indicating a wider spread of body weight outcomes in this group.

BMI by Physical Activity Status

Expectation: Physically active individuals have lower BMI



Appendix: Physical Activity Stats and T-test

Table 3.3: Descriptive Statistics for Physical Active

PhysActive	n	Mean_BMI	SD_BMI	Median_BMI
No	989	29.2	6.9	28.0
Yes	1071	27.4	5.3	26.6

Welch Two Sample t-test

data: BMI by PhysActive

t = 6.6975, df = 1847.8, p-value = 2.805e-11

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

1.285537 2.350209

sample estimates:

mean in group No mean in group Yes

29.22283

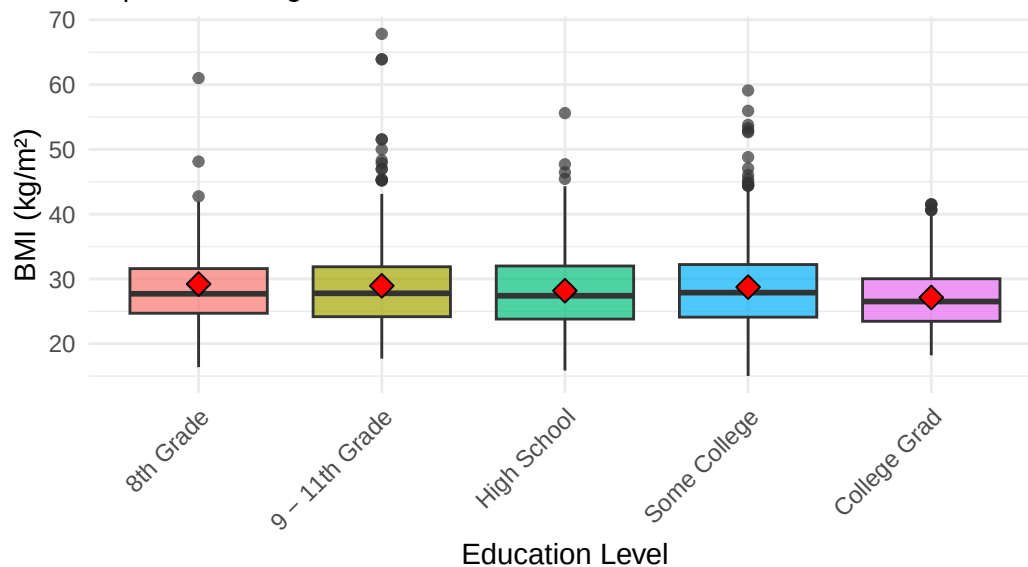
27.40496

BMI vs Education

In this sample, individuals who reported having a College Graduate had a lower mean (mean 27.1 kg/m²) compared with rest of the groups (mean 28.2 - 29.2 kg/m²). The one-way ANOVA test provides strong evidence that the mean BMI differs across education levels ($p < 0.001$). However the coefficient of determination $R^2 = 0.0126791$ indicates that the education level only explains 1.3% in the BMI variation. This means that, while the difference is statistically significant, its practical importance is very small.

BMI by Education Level

Expectation: Higher education → lower BMI



Appendix: Education Stats and Anova and

Table 3.4: Descriptive Statistics for Education

Education	n	Mean_BMI	SD_BMI	Median_BMI
8th Grade	94	29.2	6.9	27.7
9 - 11th Grade	299	28.9	7.6	27.8
Some College	690	28.7	6.2	27.9
High School	491	28.2	6.0	27.4
College Grad	486	27.1	4.9	26.5

Anova

```

              Df Sum Sq Mean Sq F value    Pr(>F)
Education      4    990   247.4    6.598 2.84e-05 ***
Residuals    2055   77068    37.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Tukey

Tukey multiple comparisons of means
95% family-wise confidence level

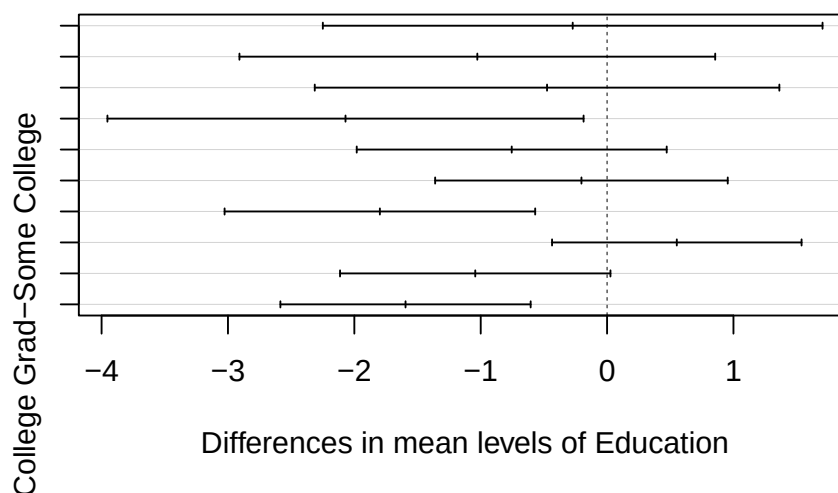
Fit: aov(formula = BMI ~ Education, data = nhanes_lm)

```

$Education
              diff      lwr      upr      p adj
9 - 11th Grade-8th Grade -0.2721042 -2.249181  1.70497290 0.9957664
High School-8th Grade    -1.0267140 -2.909063  0.85563478 0.5696549
Some College-8th Grade   -0.4749704 -2.313186  1.36324503 0.9553039
College Grad-8th Grade   -2.0699378 -3.953842 -0.18603377 0.0229035
High School-9 - 11th Grade -0.7546099 -1.981100  0.47188025 0.4467253
Some College-9 - 11th Grade -0.2028662 -1.360483  0.95475066 0.9893126
College Grad-9 - 11th Grade -1.7978337 -3.026709 -0.56895799 0.0006415
Some College-High School  0.5517436 -0.435414  1.53890127 0.5455991
College Grad-High School  -1.0432238 -2.113055  0.02660738 0.0600524
College Grad-Some College -1.5949674 -2.585087 -0.60484745 0.0001120

```

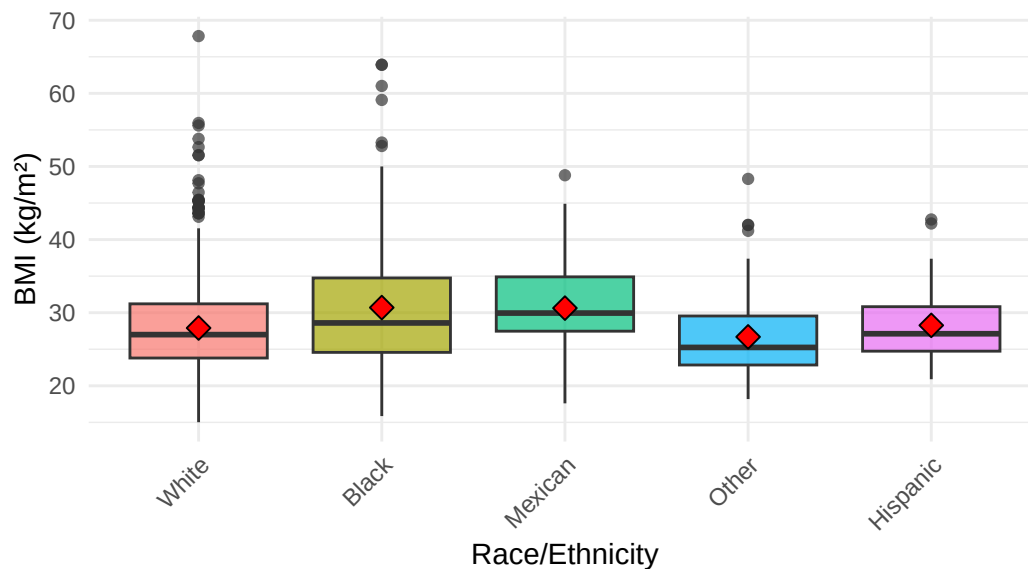
95% family-wise confidence level



BMI by Race

In this sample Black and Mexican groups show higher average BMI than White, while “Other” is lower; the boxplots (red diamonds = means) reflect these shifts.

BMI by Race



Appendinx: Race Stats and Anova

A tibble: 5 x 5

	Race1	n	Mean_BMI	SD_BMI	Median_BMI
	<fct>	<int>	<dbl>	<dbl>	<dbl>
1	Black	187	30.7	8.82	28.6
2	Mexican	125	30.6	5.72	30.0
3	Hispanic	83	28.3	4.79	27.1
4	White	1561	27.9	5.78	27
5	Other	104	26.7	5.71	25.2

Call:

```
aov(formula = BMI ~ Race1, data = nhanes_lm)
```

Terms:

	Race1	Residuals
Sum of Squares	2263.28	75794.92
Deg. of Freedom	4	2055

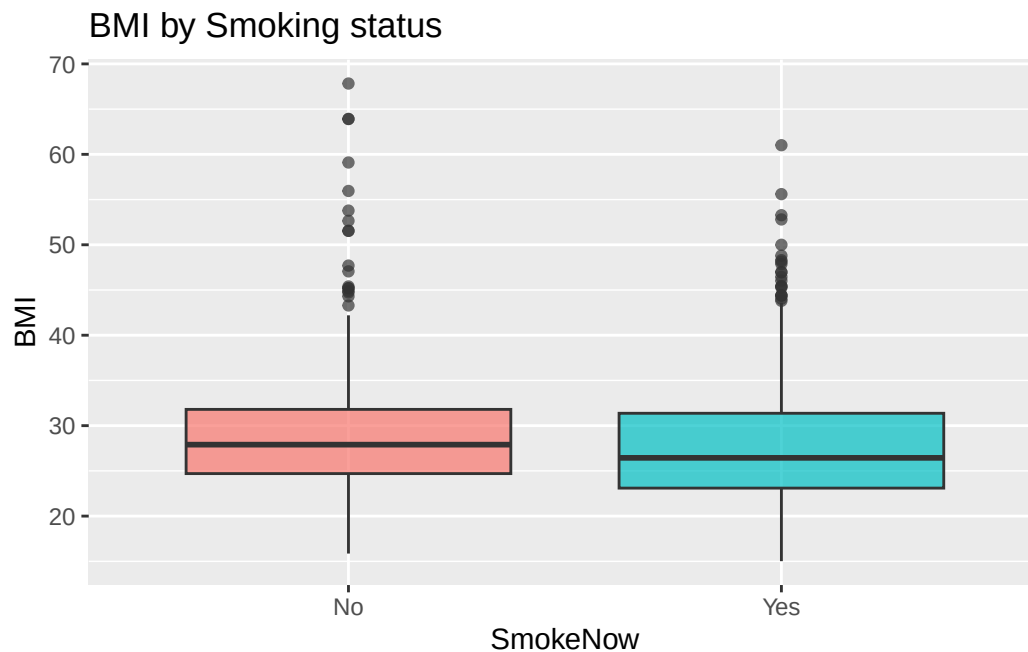
Residual standard error: 6.073152

Estimated effects may be unbalanced

One-way ANOVA indicates a significant overall difference across Race1 ($p < 0.001$); pairwise Tukey comparisons can then identify which specific pairs differ.

BMI vs Smoke

Smokers show an approximate 1.1 kg/m^2 lower mean BMI than non-smokers. The T-test result tells us that there is very strong evidence that the median BMI between non-smokers and smokers is not zero.(appendix)



Appendix: BMI~Smoke t-test

```
t_test_Smoke <- t.test(BMI ~ SmokeNow, data = nhanes_lm)
t_test_Smoke
```

Welch Two Sample t-test

data: BMI by SmokeNow

t = 4.1853, df = 1947.8, p-value = 2.974e-05

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

0.6065475 1.6762202

sample estimates:

mean in group No	mean in group Yes
28.79577	27.65439

4 Linear Model

Summary of Linear Modeling Progression (Collapsed Below)

To reach the final interaction model, we estimated a sequence of nested linear models. The simple BMI ~ Age regression showed no meaningful association, and adding basic demographics improved the fit only slightly, with race contributing the most. Socioeconomic factors provided minimal additional explanatory power. Lifestyle and clinical variables strengthened the model somewhat, with physical activity, smoking status, systolic blood pressure, and race emerging as the most consistent predictors of BMI. All earlier models and their outputs are collapsed below, while the final interaction model remains visible for interpretation.

i Linear Modeling Progression

4.1 BMI ~ Age

We are starting with a simpler linear model

The intercept is 28.01 kg/m². The interpretation has no meaning as it represents the BMI at age 0 for an adult. (part of the linear fit)

The slope is 0.005 kg/m² and the interpretation would be that for the each year increase the BMI will increase with 0.005. The p value is 0.512 and we can say that in this linear model there is no evidence of a linear association between BMI and Age.

With a fit $R^2 = 0.0002$ (adj. $R^2 = -0.0002$) Age explains none of the variability in BMI

```
m_age <- lm(BMI ~ Age, data = nhanes_lm)
```

```
#summary(m_age)
```

```
age_sum <- tidy(m_age, conf.int = TRUE)
```

```
#age_sum
```

```
age_fit <- glance(m_age)[, c("r.squared", "adj.r.squared", "sigma", "nobs")]
```

```
kable(age_sum, digits=3, caption="BMI ~ Age: coefficient table (with 95% CI)")
```

Table 4.1: BMI ~ Age: coefficient table (with 95% CI)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	28.018	0.418	67.033	0.000	27.198	28.838
Age	0.005	0.008	0.657	0.512	-0.011	0.022

```
#kable(age_fit, digits=3, caption="BMI ~ Age: model fit")
age_fit
```

```
# A tibble: 1 x 4
  r.squared adj.r.squared sigma nobs
    <dbl>         <dbl> <dbl> <int>
1  0.000209      -0.000276  6.16  2060
```

4.2 Model 1: Demographic

We will add core demographic variables to the model: Age + Gender + Race

Model: BMI ~ Age + Gender + Race

We fit a multiple linear model with BMI as the response and Age (continuous), Gender (female = reference), and Race (White = reference) as covariates.

```
M1 <- lm(BMI ~ Age + Gender + Race1, data = nhanes_lm)

summary(M1)
```

Call:

```
lm(formula = BMI ~ Age + Gender + Race1, data = nhanes_lm)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.210	-4.209	-0.949	3.436	40.186

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.117470	0.463686	58.482	< 2e-16 ***
Age	0.013510	0.008406	1.607	0.1082
Gendermale	0.221968	0.272635	0.814	0.4156
Race1Black	2.872157	0.472655	6.077	1.46e-09 ***
Race1Mexican	2.802671	0.571678	4.903	1.02e-06 ***
Race1Other	-1.131178	0.619031	-1.827	0.0678 .
Race1Hispanic	0.446401	0.688336	0.649	0.5167

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.071 on 2053 degrees of freedom

Multiple R-squared: 0.03055, Adjusted R-squared: 0.02772

F-statistic: 10.78 on 6 and 2053 DF, p-value: 7.701e-12

```
# Coefficients with 95% CIs (t-tests)
# core_coef <- broom::tidy(M1, conf.int = TRUE)
# core_coef

# M1_sum <- tidy(M1, conf.int = TRUE)
# kable(M1_sum, digits=3, caption="BMI ~ Age + Gender + Race coefficient table (with 95% CIs)")

# Model fit
# M1_fit <- broom::glance(M1)[, c("r.squared", "adj.r.squared", "sigma", "df", "nobs")]
# knitr::kable(M1_fit, digits = 3, caption = "Core model: fit statistics")
```

In this model the Race differences between groups relative to White: Black (+ 2.87 kg/m², $p < 10^{-8}$) and Mexican (+2.80 kg/m², $p < 10^{-6}$) participants have a higher BMI on average, while Other shows very weak evidence that the BMI is lower on average than White (-1.13 kg/m², $p < 0.068$) and for Hispanix group there is no evidence that the BMI is different from White category on average (+0.45 kg/m², $p < 0.517$). The adjusted $R^2 = 0.028$, meaning that demographics only explain ~2.8% of BMI variability

F-tests for M1 (BMI ~ Age + Gender + Race1)

```
M1_drop1 <- drop1(M1, test = "F")
M1_drop1
```

Single term deletions

Model:

```
BMI ~ Age + Gender + Race1
      Df Sum of Sq  RSS   AIC F value    Pr(>F)
<none>                 75673 7437.7
Age      1      95.20 75768 7438.3   2.5827    0.1082
Gender   1      24.43 75698 7436.3   0.6629    0.4156
Race1    4     2318.52 77992 7491.8  15.7252 1.109e-12 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Using drop1() we test each term conditional on the others. Race is associated with BMI; Age and Gender are not, at this stage.

4.3 Adding Socioeconomic factors

We are adding socioeconomic factors to our model

```
M2 <- update(M1, . ~ . + Education + log_income)
summary(M2)
```

Call:

```
lm(formula = BMI ~ Age + Gender + Race1 + Education + log_income,
    data = nhanes_lm)
```

Residuals:

Min	1Q	Median	3Q	Max
-14.858	-4.169	-0.913	3.454	39.780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.034963	2.066514	13.566	< 2e-16 ***
Age	0.015941	0.008473	1.881	0.0601 .
Gendermale	0.191993	0.272588	0.704	0.4813
Race1Black	2.671400	0.481595	5.547	3.28e-08 ***
Race1Mexican	2.649685	0.593333	4.466	8.41e-06 ***
Race1Other	-1.077031	0.623601	-1.727	0.0843 .
Race1Hispanic	0.344172	0.692346	0.497	0.6192
Education9 - 11th Grade	-0.030781	0.733418	-0.042	0.9665
EducationHigh School	-0.621045	0.703421	-0.883	0.3774
EducationSome College	0.164729	0.698290	0.236	0.8135
EducationCollege Grad	-1.224686	0.723380	-1.693	0.0906 .
log_income	-0.055827	0.185911	-0.300	0.7640

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.053 on 2048 degrees of freedom

Multiple R-squared: 0.03873, Adjusted R-squared: 0.03357

F-statistic: 7.501 on 11 and 2048 DF, p-value: 9.159e-13

After adding the Education and log_income as covariates the BMI remains higher for Black and Mexican participants vs White. Age and participants that are College Graduates shows a very weak association with BMI. The adj. R^2 shows an explainability of 3.4%

Term-wise F-tests

```
M2_drop1 <- drop1(M2, test = "F")
M2_drop1
```

Single term deletions

Model:

BMI ~ Age + Gender + Race1 + Education + log_income

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			75035	7430.2		
Age	1	129.68	75165	7431.8	3.5393	0.060071 .
Gender	1	18.18	75053	7428.7	0.4961	0.481305
Race1	4	1912.91	76948	7474.1	13.0527	1.687e-10 ***
Education	4	599.27	75634	7438.6	4.0891	0.002643 **
log_income	1	3.30	75038	7428.3	0.0902	0.763989

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Term-wise F-tests: BMI differs overall by Race and Education; Age is borderline; Gender and log(Income) show little added association (conditional on other covariates).

4.4 Adding Lifestyle & Clinical predictors

```
M3 <- update(M2, . ~ . + PhysActive + SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve)
summary(M3)
```

Call:

```
lm(formula = BMI ~ Age + Gender + Race1 + Education + log_income +
    PhysActive + SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve,
    data = nhanes_lm)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.915	-3.999	-0.831	3.537	37.748

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.573516	2.335666	12.662	< 2e-16 ***
Age	-0.015609	0.009670	-1.614	0.106648
Gendermale	0.014294	0.274346	0.052	0.958454
Race1Black	3.006900	0.476562	6.310	3.42e-10 ***
Race1Mexican	2.241999	0.584376	3.837	0.000129 ***
Race1Other	-0.531417	0.616321	-0.862	0.388656
Race1Hispanic	0.416809	0.679956	0.613	0.539948
Education9 - 11th Grade	0.089781	0.720029	0.125	0.900781
EducationHigh School	-0.516741	0.690459	-0.748	0.454304
EducationSome College	0.253163	0.685239	0.369	0.711830
EducationCollege Grad	-0.926147	0.719078	-1.288	0.197904
log_income	-0.110284	0.183536	-0.601	0.547982
PhysActiveYes	-1.809196	0.278270	-6.502	9.96e-11 ***
SleepHrsNight	-0.082643	0.098545	-0.839	0.401770
AlcoholDay	0.074228	0.041594	1.785	0.074476 .
SmokeNowYes	-2.181678	0.299600	-7.282	4.67e-13 ***
BPSysAve	0.022322	0.008524	2.619	0.008893 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.928 on 2043 degrees of freedom

Multiple R-squared: 0.08013, Adjusted R-squared: 0.07293

F-statistic: 11.12 on 16 and 2043 DF, p-value: < 2.2e-16

After adding lifestyle and clinical predictors to our Model BMI we can observe that BMI stays higher for Black and Mexican in comparison with White participants. Physically Active participants have a lower BMI, and Smoking participants have a lower BMI. Effects are conditional on the others (treatment coding); results are associations, not causal.

Term-wise F-tests

```
M3_drop1 <- drop1(M3, test = "F")
M3_drop1
```

Single term deletions

Model:

```
BMI ~ Age + Gender + Race1 + Education + log_income + PhysActive +  
      SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			71803	7349.5		
Age	1	91.57	71895	7350.2	2.6055	0.106648
Gender	1	0.10	71804	7347.5	0.0027	0.958454
Race1	4	1852.00	73655	7394.0	13.1736	1.346e-10 ***
Education	4	430.36	72234	7353.8	3.0612	0.015826 *
log_income	1	12.69	71816	7347.9	0.3611	0.547982
PhysActive	1	1485.64	73289	7389.7	42.2705	9.962e-11 ***
SleepHrsNight	1	24.72	71828	7348.2	0.7033	0.401770
AlcoholDay	1	111.93	71915	7350.7	3.1848	0.074476 .
SmokeNow	1	1863.69	73667	7400.3	53.0269	4.673e-13 ***
BPSysAve	1	241.01	72044	7354.4	6.8573	0.008893 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Term-wise F-tests for M3 (BMI ~ Age + Gender + Race1 + Education + log_income + PhysActive + SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve)

4.5 Adding Interactions

We extended the model with prespecified interactions to test whether the association between:

- **PhysActive** × **Gender** - Based on *Gender Differences in Exercise Habits and Quality of Life Reports*¹, physical activity patterns differ significantly by gender. Here we test whether the BMI–activity association varies by sex.
- **PhysActive** × **Education** - According to the research *Education leads to a more physically active lifestyle*², “one additional year of education leads to a 0.62-unit higher overall physical activity”. We are testing if in our sample the activity–BMI association varies across socioeconomic factors (education levels).
- **Gender** × **SmokeNow** - Based on the report from *Swiss association for tobacco control*³, there are known gender differences in smoking patterns. In our sample we are testing whether the smoking–BMI association differs by sex.

```
M4 <- update(M3, . ~ . + PhysActive:Gender + PhysActive:Education + Gender:SmokeNow)
```

i Model Summary- result

```
summary(M4)
```

Call:

¹See [PMC article](#).

²See [PMC article](#).

³See [AT report](#).

```
lm(formula = BMI ~ Age + Gender + Race1 + Education + log_income +
    PhysActive + SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve +
    Gender:PhysActive + Education:PhysActive + Gender:SmokeNow,
    data = nhanes_lm)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.168	-4.057	-0.895	3.396	37.770

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.726849	2.380324	11.648	< 2e-
16 ***				
Age	-0.014841	0.009598	-	
1.546 0.12221				
Gendermale	0.611595	0.482437	1.268	0.20504
Race1Black	2.890277	0.473409	6.105	1.23e-
09 ***				
Race1Mexican	2.292266	0.580298	3.950	8.08e-
05 ***				
Race1Other	-0.520097	0.611616	-	
0.850 0.39522				
Race1Hispanic	0.585686	0.679784	0.862	0.38902
Education9 - 11th Grade	1.154768	0.876821	1.317	0.18799
EducationHigh School	-0.145821	0.849861	-	
0.172 0.86378				
EducationSome College	0.855728	0.839405	1.019	0.30811
EducationCollege Grad	0.269230	0.937998	0.287	0.77412
log_income	-0.065774	0.183019	-	
0.359 0.71935				
PhysActiveYes	-0.262527	1.337241	-	
0.196 0.84438				
SleepHrsNight	-0.099740	0.098131	-	
1.016 0.30956				
AlcoholDay	0.084933	0.041469	2.048	0.04068 *
SmokeNowYes	-0.661655	0.433203	-	
1.527 0.12683				
BPSysAve	0.025625	0.008505	3.013	0.00262 **
Gendermale:PhysActiveYes	1.084032	0.538843	2.012	0.04437 *
Education9 - 11th Grade:PhysActiveYes	-3.183134	1.482250	-	
2.148 0.03187 *				
EducationHigh School:PhysActiveYes	-1.531381	1.409680	-	
1.086 0.27746				
EducationSome College:PhysActiveYes	-2.013071	1.376690	-	
1.462 0.14383				
EducationCollege Grad:PhysActiveYes	-2.849886	1.435170	-	
1.986 0.04720 *				
Gendermale:SmokeNowYes	-2.650774	0.538496	-4.923	9.23e-
07 ***				

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5.879 on 2037 degrees of freedom
```

```
Multiple R-squared:  0.09791,    Adjusted R-squared:  0.08817
```

```
F-statistic: 10.05 on 22 and 2037 DF,  p-value: < 2.2e-16
```

Interaction observation:

- Gender:SmokingNow: there is very strong evidence that smoking is linked to a lower BMI for both sexes: women $\sim 0.7 \text{ kg}/m^2$ lower and men $\sim 3.3 \text{ kg}/m^2$ compared with non-smokers.
- PhysActivity:Gender: there is evidence that difference in average BMI associated with physical activity is not 0 men. In the baseline group women show a small decrease with activity ($\sim 0.3 \text{ kg}/m^2$). Men add approx $1.1 \text{ kg}/m^2$ to the female difference.
- PhysActivity:Education: in lower (9–11) and higher (College) education groups, being active is associated with a noticeably lower BMI than in the 8th-grade group.

All of these interaction are conditional associations (not causal)

As we added the interaction we observe that AlcoholDay shows a small positive association with BMI (for each additional drink a day the BMI increases with $0.085 \text{ kg}/m^2$, $p = 0.04$), conditional on other covariates.

Term-wise F-tests

```
M4_drop1 <- drop1(M4, test = "F")
```

F-test result

M4_drop1

Single term deletions

Model:

BMI ~ Age + Gender + Race1 + Education + log_income + PhysActive +
SleepHrsNight + AlcoholDay + SmokeNow + BPSysAve + Gender:PhysActive +
Education:PhysActive + Gender:SmokeNow

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			70416	7321.3		
Age	1	82.64	70498	7321.7	2.3907	0.12221
Race1	4	1759.80	72175	7364.2	12.7270	3.116e-10 ***
log_income	1	4.46	70420	7319.5	0.1292	0.71935
SleepHrsNight	1	35.71	70451	7320.4	1.0330	0.30956
AlcoholDay	1	145.00	70561	7323.6	4.1947	0.04068 *
BPSysAve	1	313.77	70729	7328.5	9.0769	0.00262 **
Gender:PhysActive	1	139.91	70556	7323.4	4.0473	0.04437 *
Education:PhysActive	4	263.23	70679	7321.0	1.9037	0.10719
Gender:SmokeNow	1	837.64	71253	7343.7	24.2315	9.229e-07 ***

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Term-wise F-tests summary: BMI is associated with race, systolic BP, and shows effect modification for Gender:Smoking and Gender:Physical activity. Education:Physical activity is not supported

5 Research Questions

After adjusting for covariates, how does BMI vary with Age?

Adjusted for all covariates, the term-wise F-test (drop1) shows little evidence of a linear association between age and BMI ($p = 0.12$)

Do demographic factors show overall association with BMI?

- **Race/ethnicity:** Yes. Strong overall association (clear F-test).
- **Education:** Yes (overall main effect), but no activity–education interaction.
- **Gender:** No large main effect, but gender modifies the associations of physical activity and smoking with BMI.

Are lifestyle factors (PsyActive, AlcoholDay, SleepNight, SmokeNow) associated with BMI, and how much?

- At the reference education (8th Grade), females show a small reduction with activity (PhysActive main term).
- Males add the Gender:PhysActive interaction, show an increase with activity.
- In 9–11th Grade and College Grad, the Education:PhysActive interactions are negative, showing that there is an associated reduction than in the reference education. However with the Drop1 test there is weak to no evidence that there is an interaction between Education and PhysActive

How much variance is explained by the model?

The model's adjusted $R^2 = 0.088$, so the model explains only 8.8% of the variability in BMI.